



TalkMetrix: GenAI-Powered Customer Support Quality Auditor

Infosys Springboard Virtual Internship 6.0 | Batch 13

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The Problem & Objective

- **The Challenge:** Manual auditing of customer support (chat/voice) is slow, prone to human bias, and difficult to scale.
- **The Solution (TalkMetrix):** An automated platform leveraging Large Language Models (LLMs) to ensure quality, compliance, and effectiveness.
- **Key Goals:**
 - Automate QA scoring.
 - Detect compliance violations in real-time.
 - Provide data-driven actionable insights for BPOs and SaaS teams.

High-Level Architecture (The 'How')

- **Full-Stack Workflow:**
- **Data Ingest:** Multi-modal support (Audio & Chat logs).
- **Processing:** Whisper-based transcription for voice-to-text.
- **Intelligence:** Llama3-70B (via Groq) for deep semantic analysis.
- **Context:** RAG (Retrieval-Augmented Generation) using FAISS to check against specific company policies.
- **Output:** Dynamic dashboards and downloadable reports (PDF/Excel).

Technical Stack

- **Category Technologies Used**
- **Frontend React:** (TypeScript, Vite), Tailwind CSS, shadcn/ui
- **Backend:** FastAPI (Python), SQLite
- **AI/ML Groq:** (Llama3-70B), Whisper STT, FAISS (Vector DB)
- **Analytics Recharts:** (Data Visualization)
- **Deployment:** Netlify (Frontend), Render (Backend)

Core Features & Evaluation Metrics

- **The Scoring Engine:** Automated evaluation across four critical pillars:
- **Empathy:** Does the agent sound human and supportive?
- **Compliance:** Are legal/company protocols followed?
- **Resolution:** Was the customer's problem actually solved?
- **Professionalism:** Tone and language quality.
- **Alert System:** Instant flagging of high-risk compliance breaches.

Project Timeline & Milestones

- **Weeks 1-2:** Research & UI Foundation (React Setup).
- **Weeks 3-4:** Backend APIs & Whisper Integration.
- **Weeks 5-6:** Core Development: LLM Scoring Engine & RAG/FAISS Integration.
- **Weeks 7-8:** Analytics, Report Generation, and Production Deployment.
- **Status:** 100% Completed & Deployed.

Challenges & Engineering Solutions

- **STT Reliability:** Addressed Whisper inconsistencies by implementing robust fallback handling.
- **Prompt Engineering:** Refined prompts to ensure the LLM returned structured, parsable JSON data for the frontend.
- **Retrieval Logic:** Optimized FAISS vector search to ensure policy-based checks were fast and relevant.
- **Integration:** Managed complex API communications between the FastAPI backend and React frontend.

Visual Showcase (Screenshots)

- **Insert Dashboard View:** Highlighting the overall quality trends.
- **Insert Analytics Dashboard:** Showing agent performance charts.
- **Insert Upload/Reports Page:** Demonstrating ease of use and documentation output.

Key Learnings & Takeaways

- **Technical Growth:** Mastered the MERN/FastAPI integration and the implementation of RAG pipelines.
- **AI Integration:** Gained hands-on experience with Groq's high-speed inference and Whisper's transcription capabilities.
- **Professional Skills:** Applied Agile practices to meet tight 8-week deadlines and improved documentation standards.

Conclusion & Acknowledgments

- **Summary:** TalkMetrix successfully bridges the gap between manual QA and AI-driven efficiency.
- **Gratitude:** Thank you to the Infosys Springboard team and mentors for the guidance during Virtual Internship 6.0.
- **Q&A:** Open for questions.